
NASH BARGAINING PREFERENCE OPTIMIZATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Aligning Large Language Models (LLMs) often requires satisfying multiple, potentially competing preference objectives. This challenge has motivated growing line of work on multi-objective alignment. Many existing methods represent each objective with a scalar reward or utility, which cannot capture cyclic pairwise preferences. Game-theoretic methods such as Nash Learning from Human Feedback (NLHF) directly optimize general pairwise preferences. A recent game-theoretic approach extends this perspective to multi-objective alignment, but resolves trade-offs through worst-objective optimization. However, worst-objective optimization is only one possible way to aggregate multiple objectives. A max-min rule can admit Pareto-dominated solutions, and its solution can depend on the relative scales of the objectives. We therefore introduce Nash Bargaining Preference Optimization (NBPO), which applies Nash bargaining to objective-wise game values. When joint improvement is feasible, NBPO strictly improves every objective over a reference fallback, is Pareto optimal in game-value space, and is invariant to independent positive affine transformations of the objective utilities. For implementation, we derive a training-free regularized opponent model, and a partition-free regression loss using binary comparisons. We also establish last-iterate convergence. **Add experiment results.**

1 INTRODUCTION

Large Language Models (LLMs) are commonly refined through preference-based post-training such as Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022). Standard RLHF learns a scalar reward from pairwise comparisons and optimizes its expectation, while Direct Preference Optimization removes the separately trained reward model but retains an implicit scalar reward and a Bradley–Terry (BT) representation of preferences (Rafailov et al., 2023; Bradley & Terry, 1952).

A deployed assistant is evaluated along several objectives, including helpfulness, harmlessness, correctness, and conciseness. Existing multi-objective methods expose or constrain their trade-offs through objective weights, policy interpolation, preference-conditioned policies, hard constraints, or worst-case aggregation (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These approaches are useful when users can specify weights, thresholds, a primary objective, or a weakest-objective rule. We instead study the complementary setting in which a fixed set of objectives are combined into one deployed policy without prespecified priorities.

General pairwise feedback creates a separate difficulty. Scalar-reward models cannot represent a cycle in which response a is preferred to b , b to c , and c to a . Nash Learning from Human Feedback (NLHF) treats such feedback directly as a zero-sum preference game, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Multi-objective extensions use Blackwell target sets or a maximum-entropy worst-objective game (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026). By contrast, fixed-reference methods such as MOPO compare each objective only against a fixed behavior distribution, which is computationally convenient but can discard pairwise structure that is invisible from that reference (Agnihotri et al., 2026; Zhang et al., 2026).

These strands leave two conceptually different decisions entangled: how to summarize possibly cyclic preferences *within* each objective, and how to aggregate the resulting values *across* objectives. A maximin rule answers both questions at once by placing objective weights and opponent policies inside a single worst-case game. We instead preserve one game value per objective and then apply an explicit social aggregation rule to the resulting vector.

We introduce Nash Bargaining Preference Optimization (NBPO). For objective k , an adaptive opponent policy searches for responses that expose weaknesses of the current policy, while a KL penalty keeps that opponent near the deployed reference policy. This yields an entropy-regularized game value that depends on the full pairwise comparison matrix. The deployed reference also supplies the bargaining fallback: the improvement for objective k is the learned policy’s game value minus the reference policy’s game value. NBPO maximizes the sum of the logarithms of these positive improvements. The inner preference game and the outer Nash bargaining rule therefore play distinct roles.

This separation gives a clear set of guarantees. When one policy can improve every objective over the reference, every NBPO optimizer is strictly above the fallback in all game-value coordinates and Pareto optimal in those coordinates. Although several policies may realize the same outcome, the selected game-value vector is unique. The solution is proportionally fair and unaffected by independent positive affine changes of objective utility scales. Utilitarian (Averaging) and maximin rules do not satisfy this complete set of properties without additional tie-breaking or fallback constraints.

The algorithm follows the same separation. A feasibility phase first enters a statistically certified positive-improvement region. The bargaining phase then combines objective-specific opponent distributions with inverse-surplus weights, and implements the entropy-mirror step through pairwise log-ratio regression without fitting scalar rewards. We provide simultaneous confidence intervals for the nested game values and one last-iterate bound that explicitly includes opponent-estimation and neural-regression errors. In the available matched finite-temperature run on HELPSTEER2 and SAFERLHF, all methods have positive reported objective improvements, but NBPO does not uniformly outperform fixed aggregation rules. We therefore treat the experiments as evidence for the implementation and diagnostics, not as a blanket performance claim.

Our contributions are:

- We formulate one-policy multi-objective alignment as Nash bargaining over objective-wise game values derived directly from general pairwise preference oracles.
- We prove strict fallback-relative improvement, Pareto optimality, uniqueness of the selected game-value vector, proportional fairness, and positive-affine covariance under joint improvement, and contrast these guarantees with utilitarian and maximin aggregation.
- We derive a closed-form regularized opponent, a certified warm start, partition-free pairwise regression, and a unified last-iterate guarantee that covers exact and approximate updates.

2 BACKGROUND

General pairwise preference oracles. A prompt x is drawn from ρ , and a policy $\pi(\cdot | x)$ is a distribution over a finite response set $\mathcal{Y}_x$. Objective $k \in [K]$ is represented by a pairwise preference oracle $P_k(y \succ z | x) \in [0, 1]$ satisfying

$$P_k(y \succ z | x) + P_k(z \succ y | x) = 1, \quad P_k(y \succ y | x) = \frac{1}{2}. \quad (1)$$

Define the centered payoff

$$A_k(y, z | x) = P_k(y \succ z | x) - \frac{1}{2}, \quad A_k(y, z | x) = -A_k(z, y | x), \quad (2)$$

and, for policies π and ν ,

$$g_k(\pi, \nu) = \mathbb{E}_{x \sim \rho, y \sim \pi(\cdot | x), z \sim \nu(\cdot | x)} A_k(y, z | x). \quad (3)$$

No scalar reward representation or transitivity assumption is imposed. We write the contextual policy inner product as $\langle q, \pi \rangle_\rho = \mathbb{E}_x \sum_y \pi(y | x) q(y | x)$ and the contextual KL analogously.

Bargaining problems. A K -dimensional bargaining problem is a feasible utility set $\mathcal{G} \subseteq \mathbb{R}^K$ and disagreement point $d \in \mathcal{G}$. It is nondegenerate if some $u \in \mathcal{G}$ satisfies $u > d$ componentwise. On a compact, convex, comprehensive problem, the Nash solution maximizes

$$\sum_{k=1}^K \log(u_k - d_k) \quad (4)$$

over utilities strictly above disagreement (Nash, 1950). Its derivative in coordinate k is the inverse surplus $1/(u_k - d_k)$.

3 OBJECTIVE-WISE PREFERENCE GAMES

This section turns each objective-wise preference oracle into a scalar policy utility by defining an entropy-regularized preference game, deriving its adaptive opponent and policy gradient, and illustrating why this utility captures cyclic structure that fixed-reference evaluation can miss. The reference policy μ has two distinct roles. It is the deployed fallback used by the bargaining problem, and it anchors the opponent distribution used to evaluate each objective. Sharing the same policy does not merge these roles.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k,\beta_k}(\pi) = \min_{\nu} \{g_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \rho} \text{KL}(\nu(\cdot | x) \| \mu(\cdot | x))\}. \quad (5)$$

The opponent ν searches for responses that perform well against π under objective k . The KL term restricts this search to a neighborhood of μ . Thus, a large β_k keeps the opponent close to μ , whereas a small β_k gives more weight to worst-case comparisons. The limit $\beta_k \downarrow 0$ is the unregularized worst-case game value $\min_{\nu} g_k(\pi, \nu)$.

Let

$$r_{k,\pi}(z | x) = \mathbb{E}_{y \sim \pi(\cdot | x)} A_k(y, z | x). \quad (6)$$

The minimizing opponent and the game value are

$$\nu_{k,\pi}^*(z | x) = \frac{\mu(z | x) \exp\{-r_{k,\pi}(z | x)/\beta_k\}}{\mathbb{E}_{z' \sim \mu(\cdot | x)} \exp\{-r_{k,\pi}(z' | x)/\beta_k\}}, \quad (7)$$

$$V_{k,\beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \rho} \log \mathbb{E}_{z \sim \mu(\cdot | x)} \exp\{-r_{k,\pi}(z | x)/\beta_k\}. \quad (8)$$

Proposition 1 (Structure). *For every k and $\beta_k \geq 0$, V_{k,β_k} is continuous and concave in π . For $\beta_k > 0$ it is differentiable, with*

$$q_{k,\pi}(y | x) = \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} A_k(y, z | x), \quad DV_{k,\beta_k}(\pi)[h] = \langle q_{k,\pi}, h \rangle_{\rho}, \quad (9)$$

where D is the directional derivative operator. The adaptive opponent makes $V_{k,\beta_k}(\pi)$ depend on the full pairwise comparison matrix, rather than only on comparisons with one fixed reference response.

Why a fixed reference can miss cyclic structure. Consider one prompt and one objective. Let δ_y denote the deterministic policy that always outputs response y , and let the reference be $\mu = \delta_r$. Suppose A , B , and C each beat r with probability 0.75, but form the deterministic cycle $A \succ B$, $B \succ C$, and $C \succ A$. The pure policy δ_A and the uniform mixture $\pi_{\text{mix}} = (\delta_A + \delta_B + \delta_C)/3$ have the same fixed-reference score, $P(\pi \succ r) - 1/2 = 0.25$. Their worst-case game values differ: at $\beta = 0$, response C defeats δ_A , so $V_0(\delta_A) = -0.5$, whereas the uniform mixture ties every response in the cycle, so $V_0(\pi_{\text{mix}}) = 0$. Since $V_0(\mu) = -0.25$, their improvements over the fallback are -0.25 and 0.25 , respectively. A fixed reference therefore cannot distinguish a fully exploitable pure response from a balanced mixture once both have the same reference win rate. Appendix A.2 gives the full calculation and formalizes the limiting relation to the fixed-reference precursor.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

This section converts the objective-wise game values from Section 3 into a bargaining problem over improvements relative to the reference. We state the feasibility condition under which this problem is

well defined, characterize the resulting Nash solution, and compare its population guarantees with utilitarian and maxmin aggregation. The reference defines one fallback value and one improvement (the bargaining *surplus*) per objective:

$$d_k = V_{k,\beta_k}(\mu), \quad s_k(\pi) = V_{k,\beta_k}(\pi) - d_k. \quad (10)$$

A positive surplus means that π has a higher regularized game value than the deployed reference for objective k . Because the Nash objective requires every surplus to be positive, we first state the feasibility and support conditions under which the bargaining problem is well defined.

Assumption 1 (Joint improvement and support). *The reference μ has full support, and some policy $\bar{\pi}$ satisfies $s_k(\bar{\pi}) > 0$ for all k .*

The assumption can fail when the reference is already optimal for one objective or when no policy jointly improves the full objective set.

On $\Pi_+ = \{\pi : s_k(\pi) > 0 \forall k\}$, NBPO selects

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) = \sum_{k=1}^K \log s_k(\pi). \quad (11)$$

We do not add an objective-level KL term. KL appears only in the opponent definition and as the proximal geometry used to optimize the same objective.

Let $V_\beta(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi))$ and define the comprehensive individually rational hull

$$\mathcal{G}_\beta = \{u \in \mathbb{R}^K : d \leq u \leq V_\beta(\pi) \text{ componentwise for some } \pi\}. \quad (12)$$

The next result verifies that these policy-induced game values form a valid classical bargaining problem; this is the step that justifies applying the Nash solution and its standard outcome guarantees.

Theorem 1 (Bargaining outcome). *Under Assumption 1, $\mathcal{G}_\beta$ is compact, convex, comprehensive, and nondegenerate. Maximizing (11) is equivalent to applying the classical Nash solution to $(\mathcal{G}_\beta, d)$. Every optimizer strictly improves every objective over the fallback and is Pareto optimal in V_β , and all optimizers induce the same game-value vector.*

The policy itself can remain nonunique because distinct response distributions may realize the same game-value vector. Beyond these core outcome guarantees, two further properties are especially relevant when objectives use different numerical scales and when improvements must be balanced across judges.

Theorem 2 (Scale covariance). *For arbitrary $c_k > 0$ and $b_k \in \mathbb{R}$, replacing (V_k, d_k) by $(c_k V_k + b_k, c_k d_k + b_k)$ leaves the policy optimizer set unchanged.*

To clarify why these guarantees favor Nash bargaining over common aggregation alternatives, Table 1 compares their population-level properties under the same joint-positive-surplus setting. Each entry is a guarantee for every global population optimizer; positive-weight utilitarian uses fixed weights $w_k > 0$, and affine covariance refers to jointly transforming each game-value coordinate and its fallback.

Table 1: Population guarantees under joint positive surplus.

rule	strict IR	Pareto optimal	unique value	scale covariance
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

Appendix A.5 gives explicit finite-game counterexamples for the other rules. Appendix A.4 also distinguishes covariance of the utility representation from rescaling raw preference labels at fixed temperature.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

This section turns the Nash objective in Section 4 into a trainable preference-optimization procedure. We first certify entry into the positive-surplus region, then implement inverse-surplus mirror ascent with partition-free binary regression, and finally establish last-iterate convergence.

5.1 POPULATION GRADIENT AND FEASIBILITY PHASE

By Proposition 1, the Nash gradient is

$$\nabla F(\pi_t) = \sum_{k=1}^K \frac{q_{k,\pi_t}}{s_k(\pi_t)}, \quad q_{k,\pi_t}(y | x) = \mathbb{E}_{z \sim \nu_{k,\pi_t}^*} A_k(y, z | x). \quad (13)$$

Thus NBPO combines two endogenous mechanisms. The regularized opponent ν_{k,π_t}^* identifies responses that exploit the current policy under objective k , while $1/s_k(\pi_t)$ allocates greater marginal influence to objectives with smaller fallback-relative improvement.

Because $s_k(\mu) = 0$, training begins with a feasibility phase. Define

$$H_{\tau_0}(\pi) = -\tau_0 \log \sum_{k=1}^K \exp\{-s_k(\pi)/\tau_0\}, \quad (14)$$

which is a concave smooth lower approximation to $\min_k s_k(\pi)$. Its gradient uses softmin objective weights and the same game gradients $q_{k,\pi}$. Phase 0 stops only after every held-out surplus has a positive lower confidence bound.

A valid game-value certificate. The game value in (8) contains a nested expectation and cannot be certified by applying a one-level Hoeffding bound directly to pairwise wins. Appendix B gives an explicit nested-sampling construction. For each checked policy it draws n independent prompts, M reference comparators per prompt, and L independent learner comparisons per comparator. It returns simultaneous intervals

$$\underline{V}_k(\pi) \leq V_{k,\beta_k}(\pi) \leq \bar{V}_k(\pi) \quad (15)$$

for all objectives and all predeclared checks with probability at least $1 - \delta$. The valid surplus interval is

$$\underline{s}_k(\pi) = \underline{V}_k(\pi) - \bar{V}_k(\mu), \quad \bar{s}_k(\pi) = \bar{V}_k(\pi) - \underline{V}_k(\mu). \quad (16)$$

An uncertified Phase 0 return does not prove population infeasibility. It means that the chosen optimization and evaluation budget did not establish joint positive surplus.

5.2 MIRROR ASCENT AND PARTITION-FREE REGRESSION

Once $\underline{s}_k(\pi_t) > 0$ for every objective, define normalized inverse-surplus weights

$$\bar{\omega}_{t,k} = \frac{1/\hat{s}_k(\pi_t)}{\sum_j 1/\hat{s}_j(\pi_t)}, \quad W_t = \sum_j \frac{1}{\hat{s}_j(\pi_t)}. \quad (17)$$

Normalization changes only the effective step size. With exact values, the entropy-mirror population update is

$$\pi_{t+1} = \arg \max_{\pi} \{\alpha_t \langle \nabla F(\pi_t), \pi \rangle_{\rho} - \mathbb{E}_x \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))\}, \quad (18)$$

where α_t is the step size at t , and closed form is

$$\pi_{t+1}(y | x) \propto \pi_t(y | x) \exp \left\{ \alpha_t \sum_{k=1}^K \frac{\mathbb{E}_{z \sim \nu_{k,\pi_t}^*} A_k(y, z | x)}{s_k(\pi_t)} \right\}. \quad (19)$$

For $y, y' \sim \pi_t(\cdot | x)$ and $z_k \sim \nu_{k,\pi_t}^*(\cdot | x)$, query independent binary outcomes $B_k(y, z_k), B_k(y', z_k) \in \{0, 1\}$ and set

$$Z_{t,k} = B_k(y, z_k) - B_k(y', z_k). \quad (20)$$

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference μ , preference oracles $P_{1:K}$, temperatures $\beta_{1:K}$, floor ε , confidence δ , coefficients η_t .

- 1: Estimate and cache simultaneous intervals for $d_k = V_{k, \beta_k}(\mu)$.
- 2: From μ , optimize the softmin feasibility objective (14).
- 3: **if** no checkpoint has $s_k \geq 2\varepsilon$ for every k **then**
- 4: **return** “no certified joint game-value improvement”.
- 5: **end if**
- 6: Set the certified checkpoint to π_0 .
- 7: **for** $t = 0, \dots, T - 1$ **do**
- 8: **for** $k = 1, \dots, K$ **do**
- 9: Estimate r_{k, π_t} , sample the regularized opponent by (7), and estimate $s_k(\pi_t)$ on the fresh or preallocated certification batch for this check.
- 10: **end for**
- 11: Form normalized inverse-surplus weights (17).
- 12: Fit a candidate $\tilde{\pi}_{t+1}$ with the regression loss (22).
- 13: Compute simultaneous surplus intervals and a lower confidence bound on $F(\tilde{\pi}_{t+1}) - F(\pi_t)$.
- 14: **if** $\min_k s_k(\tilde{\pi}_{t+1}) < \varepsilon$ **then**
- 15: Shrink η_t and repeat, or retain π_t if the stage budget is exhausted.
- 16: **else if** the welfare-improvement lower bound is negative **then**
- 17: Increase the paired certification budget if available; otherwise retain π_t .
- 18: **else**
- 19: $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$.
- 20: **end if**
- 21: **end for**
- 22: **return** π_T .

Conditionally, $\mathbb{E}[Z_{t,k} \mid y, y', z_k] = A_k(y, z_k) - A_k(y', z_k)$. Let

$$h_t(\pi; y, y') = \log \frac{\pi(y \mid x)}{\pi_t(y \mid x)} - \log \frac{\pi(y' \mid x)}{\pi_t(y' \mid x)}. \quad (21)$$

The population loss

$$\mathbb{E} \left[h_t(\pi; y, y') - \eta_t \sum_{k=1}^K \bar{\omega}_{t,k} Z_{t,k} \right]^2 \quad (22)$$

has the normalized mirror update as its unique full-support minimizer, with effective step $\alpha_t = \eta_t / W_t$. The objective-specific opponents are sampled by importance reweighting reference responses using (7); no opponent language model is trained.

5.3 LAST-ITERATE CONVERGENCE

Let $\Pi_\varepsilon = \{\pi : \min_k s_k(\pi) \geq \varepsilon\}$. On this region, F is smooth relative to negative entropy with

$$L_\varepsilon = \frac{1}{4} \sum_{k=1}^K \left(\frac{1}{\beta_k \varepsilon} + \frac{1}{\varepsilon^2} \right). \quad (23)$$

Theorem 3 (Exact last-iterate welfare and surplus convergence). *Let π^* maximize (11), and let $\{\pi_t\}_{t=0}^T$ be the exact full-support mirror iterates of (18) with a constant effective step $\alpha_t \equiv \alpha$. Suppose $\pi^*, \pi_0, \dots, \pi_T \in \Pi_\varepsilon$ and $0 < \alpha \leq 1/L_\varepsilon$. Then $F(\pi_t)$ is nondecreasing and*

$$F(\pi^*) - F(\pi_T) \leq \frac{\mathbb{E}_x \text{KL}(\pi^* \parallel \pi_0)}{\alpha T}. \quad (24)$$

If s^* is the unique optimal surplus vector, then

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2\mathbb{E}_x \text{KL}(\pi^* \parallel \pi_0)}{\alpha T}. \quad (25)$$

Thus Theorem 3 gives $O(1/T)$ last-iterate welfare convergence and $O(T^{-1/2})$ convergence of the surplus vector. Appendix A.7 gives the variable-step and approximate fitted-update extension, where gradient-estimation and mirror-step errors appear explicitly.

Table 2: LLM as a judge results. All entries are swap-averaged win rates against the base policy on the benchmark’s own prompts, judged by a single pinned open-weights judge. These are open-judge pairwise proxies rather than canonical GPT scores. “Alpaca LC” is the length-controlled AlpacaEval variant, “len.-matched” the Arena-Hard win rate restricted to response pairs within a $1.3\times$ length band, and “mean tokens” the mean generated Arena-Hard length; the last two are included because open judges reward verbosity. MOPO is reported at both temperatures the original paper sweeps.

method	Arena-Hard v2	AlpacaEval 2	Alpaca LC	MT-Bench	len.-matched	mean tokens
Certified warm start (Phase 0)	0.508	0.517	0.518	0.513	0.514	2859
NBPO (ours)	0.516	0.533	0.532	0.503	0.520	2940
NBPO, prompt-wise	0.513	0.543	0.535	0.531	0.528	2986
PROSPER/MaxEntBW	0.515	0.546	0.541	0.560	0.526	3124
Utilitarian	0.522	0.544	0.548	0.516	0.529	3045
Absolute maxmin	0.532	0.536	0.548	0.525	0.529	3130
Surplus maxmin	0.505	0.532	0.524	0.503	0.536	2978
Uniform-INPO	0.527	0.517	0.535	0.503	0.535	2976
MOPO ($\tau=0.1$)	0.508	0.512	0.516	0.513	0.495	2862
MOPO ($\tau=1$)	0.511	0.496	0.512	0.466	0.526	2829

6 EXPERIMENTS

Setup. We evaluate four objectives—helpfulness, honesty, instruction following, and truthfulness—on ULTRAFEEDBACK. The base and reference policy are Llama-3.1-8B-Instruct. A prompted Qwen3-32B preference oracle supplies objective-specific training comparisons; each pair is queried in both presentation orders and swap-averaged. Final evaluation uses Llama-3.3-70B. Appendix H gives the rubrics and label construction.

Results.

7 RELATED WORK

General pairwise and multi-objective preferences. NLHF and its scalable variants optimize directly against general pairwise preference oracles rather than fitting a scalar reward (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy regression update and an inner worst-case choice over objective weights and opponent policies (Zhang et al., 2026). NBPO instead retains one regularized game value per objective and applies reference-relative Nash bargaining across those values. The contribution is this separation of within-objective preference representation from across-objective aggregation, together with the corresponding optimization, certification, and convergence analysis.

Multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize a weakest-objective rule, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These approaches provide useful control when weights, priorities, or thresholds are available. NBPO addresses a different setting: one compromise policy for a fixed objective panel without prespecified priorities. RACO applies clipped conflict-averse gradient combination to objective-specific preference losses and traces a Pareto frontier for user-specified weights (Chen et al., 2026). It is therefore a relevant frontier method, but does not select a single priority-free compromise.

Social choice and bargaining. Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards, and separately considers reward-free von Neumann solutions (Zhong et al., 2024a). Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our setting applies the Nash rule to improvements in objective-wise game values derived directly from separate pairwise preference oracles.

8 CONCLUSION AND LIMITATIONS

NBPO separates two decisions in multi-objective alignment: an objective-wise preference game summarizes potentially cyclic pairwise feedback, and Nash bargaining selects one compromise across the resulting game values. Under joint improvement, the selected outcome is strictly above the reference in every objective, Pareto optimal, unique in game-value space, proportionally fair, and covariant to independent positive affine utility transformations. A certified warm start, regularized opponent sampling, and pairwise regression yield a practical optimizer, while Theorem 3 exposes both population and neural-update errors in one statement. Appendix A.2 records the fixed-reference precursor as a limiting case rather than making it a main contribution.

The guarantees apply to the selected game-value coordinates, not to every prompt, every pairwise comparison, or an external safety metric. The objective panel, reference policy, and opponent temperature remain modeling choices, and joint positive improvement can be infeasible. Worst-opponent evaluation can favor conservative mixtures and increases sampling cost. Finally, the current completed comparison contains one matched training run per method; the observed ordering must be replicated across independent seeds before it can support a performance claim.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 specifies the training procedure. Appendices A–B provide the population proofs, finite-sample certificate, and approximate-update analysis. Appendix E gives detailed results and the matched evaluation protocol. A submission package should include anonymized code, exact prompts, random seeds, model checkpoints, opponent pools, and per-stage confidence intervals.

REFERENCES

- Akhil Agnihotri, Rahul Jain, Deepak Ramachandran, and Zheng Wen. Multi-objective preference optimization: Improving human alignment of generative models. In *International Conference on Machine Learning*, 2026.
- Kush Bhatia, Ashwin Pananjady, Peter L. Bartlett, Anca D. Dragan, and Martin J. Wainwright. Preference learning along multiple criteria: A game-theoretic perspective. In *Advances in Neural Information Processing Systems*, volume 33, pp. 7413–7424, 2020.
- David Blackwell. An analog of the minimax theorem for vector payoffs. *Pacific Journal of Mathematics*, 6(1):1–8, 1956.
- Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Daniele Calandriello, Daniel Guo, Rémi Munos, Mark Rowland, Yunhao Tang, Bernardo Avila Pires, Pierre Harvey Richemond, Charline Le Lan, Michal Valko, Tianqi Liu, et al. Human alignment of large language models through online preference optimisation. *arXiv preprint arXiv:2403.08635*, 2024.

-
- Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- Peter Chen, Xiaopeng Li, Xi Chen, and Tianyi Lin. Reward-free alignment for conflicting objectives. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026.
- Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Vincent Conitzer, Rachel Freedman, Jobst Heitzig, Wesley H. Holliday, Bob M. Jacobs, Nathan Lambert, Milan Mossé, Eric Pacuit, Stuart Russell, Hailey Schoelkopf, Emanuel Tewolde, and William S. Zwicker. Position: Social choice should guide AI alignment in dealing with diverse human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.
- Rémi Munos, Michal Valko, Daniele Calandriello, Mohammad Gheshlaghi Azar, Mark Rowland, Daniel Guo, Yunhao Tang, Matthieu Geist, Thomas Mesnard, Andrea Michi, et al. Nash learning from human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- John F Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.
- Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-task learning as a bargaining game. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2022.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pp. 27730–27744, 2022.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Alexandre Ramé, Guillaume Couairon, Corentin Dancette, Jean-Baptiste Gaya, Mustafa Shukor, Laure Soulier, and Matthieu Cord. Rewarded soups: Towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Corby Rosset, Ching-An Cheng, Arindam Mitra, Michael Santacroce, Ahmed Awadallah, and Tengyang Xie. Direct nash optimization: Teaching language models to self-improve with general preferences. *arXiv preprint arXiv:2404.03715*, 2024.
- Seongho Son, William Bankes, Sangwoong Yoon, Shyam Sundhar Ramesh, Xiaohang Tang, and Ilija Bogunovic. Robust multi-objective controlled decoding of large language models. In *International Conference on Learning Representations*, 2026.
- Yue Wu, Zhiqing Sun, Huizhuo Yuan, Kaixuan Ji, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment. In *International Conference on Learning Representations*, 2025.
- Jiahao Zhang, Lujing Zhang, Keltin Grimes, Zhuohao Yu, Gokul Swamy, and Zhiwei Steven Wu. Back to blackwell: Closing the loop on intransitivity in multi-objective preference fine-tuning. *arXiv preprint arXiv:2602.19041*, 2026.

Yuheng Zhang, Dian Yu, Baolin Peng, Linfeng Song, Ye Tian, Mingyue Huo, Nan Jiang, Haitao Mi, and Dong Yu. Iterative nash policy optimization: Aligning LLMs with general preferences via no-regret learning. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=Pujt3ADZgI>.

Huiying Zhong, Zhun Deng, Weijie J. Su, Zhiwei Steven Wu, and Linjun Zhang. Provable multi-party reinforcement learning with diverse human feedback. *arXiv preprint arXiv:2403.05006*, 2024a.

Yifan Zhong, Chengdong Ma, Xiaoyuan Zhang, Ziran Yang, Haojun Chen, Qingfu Zhang, Siyuan Qi, and Yaodong Yang. Panacea: Pareto alignment via preference adaptation for LLMs. In *Advances in Neural Information Processing Systems*, 2024b.

Runlong Zhou, Maryam Fazel, and Simon S. Du. Extragradients preference optimization (EGPO): Beyond last-iterate convergence for nash learning from human feedback. In *Conference on Language Modeling*, 2025.

Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

A PROOFS AND ADDITIONAL THEORY

Throughout the proofs, response sets are finite. The contextual case uses the ρ -weighted product of policy simplices and the ρ -weighted KL divergence. We therefore write most curvature calculations for one prompt; the contextual extension is block diagonal.

A.1 PROOF OF PROPOSITION 1

For fixed ν , the objective in (5) is affine in π . A pointwise infimum of affine functions is concave. Compactness of the finite policy simplices and continuity of the payoff imply continuity.

For $\beta_k > 0$, the minimization separates over prompts. Fix x and abbreviate $r(z) = r_{k,\pi}(z | x)$. The Lagrangian for

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) r(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\mu(z)}$$

gives $\log(\nu(z)/\mu(z)) = -r(z)/\beta_k - c_x$. Normalization yields (7), and substitution gives (8). The minimizer is unique because KL is strictly convex on the support of μ . Danskin’s theorem therefore gives

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_x \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} A_k(y, z | x),$$

where $z \sim \nu_{k,\pi}^*$, which is (9).

A.2 CONNECTION TO THE FIXED-REFERENCE PRECURSOR

Define the fixed-reference improvement

$$a_k(\pi) = g_k(\pi, \mu) = P_k(\pi \succ \mu) - \frac{1}{2}. \quad (26)$$

Skew-symmetry gives $a_k(\mu) = 0$.

Full cyclic example. For one prompt, let $\mu = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle $A \succ B$, $B \succ C$, and $C \succ A$. Both $\pi_{\text{pure}} = \delta_A$ and $\pi_{\text{mix}} = (\delta_A + \delta_B + \delta_C)/3$ have fixed-reference improvement $0.75 - 0.5 = 0.25$. At temperature zero, however, C defeats π_{pure} deterministically, so $V_0(\pi_{\text{pure}}) = -0.5$. Against each of A, B, C , the mixture wins with probability $(1/2 + 0 + 1)/3 = 1/2$, and it beats r with probability 0.75, so $V_0(\pi_{\text{mix}}) = 0$. The reference loses to each cycle response with probability 0.75, giving $V_0(\mu) = 0.25 - 0.5 = -0.25$. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\mu)$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference collapses two policies with different exploitability to the same score; the objective-wise game value retains the distinction.

Theorem 4 (Fixed-reference and worst-opponent limits). *For every policy π and $\beta_k > 0$,*

$$0 \leq a_k(\pi) - V_{k,\beta_k}(\pi) \leq \frac{1}{8\beta_k}. \quad (27)$$

If $d_{k,\beta_k} = V_{k,\beta_k}(\mu)$ and $s_{k,\beta_k}(\pi) = V_{k,\beta_k}(\pi) - d_{k,\beta_k}$, then

$$\sup_{\pi} |s_{k,\beta_k}(\pi) - a_k(\pi)| \leq \frac{1}{8\beta_k}. \quad (28)$$

Consequently, $s_{k,\beta_k}(\pi) \rightarrow a_k(\pi)$ uniformly as $\beta_k \rightarrow \infty$. If μ has full support on the finite response set, then

$$V_{k,\beta_k}(\pi) \longrightarrow \min_{\nu} g_k(\pi, \nu) \quad \text{as } \beta_k \downarrow 0. \quad (29)$$

Corollary 1 (Consistency with the fixed-reference precursor). *Suppose some policy satisfies $a_k(\pi) > 0$ for every objective. For any sequence of temperature vectors with $\min_k \beta_k \rightarrow \infty$, every accumulation point of finite-temperature Nash optimizers maximizes $\sum_k \log a_k(\pi)$. Their fixed-reference improvement vectors converge to the unique utility vector selected by the precursor bargaining problem.*

A large-temperature expansion also gives

$$V_{k,\beta_k}(\pi) = a_k(\pi) - \frac{1}{2\beta_k} \mathbb{E}_x \text{Var}_{z \sim \mu} [r_{k,\pi}(z | x)] + O(\beta_k^{-2}), \quad (30)$$

so finite temperature adds a correction that depends on how unevenly the current policy performs across reference responses.

Proof of Theorem 4. Fix a prompt and let $R = r_{k,\pi}(Z | x)$ for $Z \sim \mu(\cdot | x)$. Since $A_k \in [-1/2, 1/2]$, also $R \in [-1/2, 1/2]$. Write

$$V_x = -\beta_k \log \mathbb{E} e^{-R/\beta_k}.$$

Jensen's inequality gives $\log \mathbb{E} e^{-R/\beta_k} \geq -\mathbb{E} R/\beta_k$, hence $V_x \leq \mathbb{E} R$. Hoeffding's lemma gives

$$\log \mathbb{E} \exp \left\{ -\frac{R - \mathbb{E} R}{\beta_k} \right\} \leq \frac{1}{8\beta_k^2}.$$

Therefore $V_x \geq \mathbb{E} R - 1/(8\beta_k)$. Averaging over x proves (27) because $\mathbb{E}_x \mathbb{E} R = g_k(\pi, \mu) = a_k(\pi)$.

Apply the same argument at $\pi = \mu$. Since $a_k(\mu) = g_k(\mu, \mu) = 0$, there are errors $e_{\pi}, e_{\mu} \in [0, 1/(8\beta_k)]$ such that

$$V_{k,\beta_k}(\pi) = a_k(\pi) - e_{\pi}, \quad V_{k,\beta_k}(\mu) = -e_{\mu}.$$

Thus $s_{k,\beta_k}(\pi) - a_k(\pi) = e_{\mu} - e_{\pi}$, proving (28).

For the zero-temperature limit, at each prompt

$$-\beta_k \log \sum_z \mu(z | x) e^{-r(z)/\beta_k} \longrightarrow \min_{z: \mu(z|x) > 0} r(z).$$

Full support makes this the minimum over all responses, which equals $\min_{\nu} g_k(\pi, \nu)$ after averaging over prompts.

Proof of Corollary 1. Let $\epsilon_n = \max_k (8\beta_{n,k})^{-1} \rightarrow 0$ and let $s_{n,k}$ denote the corresponding game surplus. Theorem 4 gives $\sup_{\pi} |s_{n,k}(\pi) - a_k(\pi)| \leq \epsilon_n$. Choose a policy $\bar{\pi}$ with $m = \min_k a_k(\bar{\pi}) > 0$. For all sufficiently large n , $s_{n,k}(\bar{\pi}) \geq m/2$, so an optimizer π_n satisfies

$$F_n(\pi_n) \geq K \log(m/2).$$

Every positive game surplus is at most one. Hence each coordinate obeys $\log s_{n,k}(\pi_n) \geq F_n(\pi_n)$, and therefore $s_{n,k}(\pi_n) \geq (m/2)^K$. Thus the optimizer sequence remains uniformly away from the surplus boundary. On that compact interior region, $\sum_k \log s_{n,k}$ converges uniformly to $\sum_k \log a_k$.

Compactness and the standard argmax argument imply that every accumulation point of π_n maximizes the fixed-reference objective. The fixed-reference bargaining problem has a unique utility vector, so the full sequence of anchor-surplus vectors converges to it.

Finally, the cumulant expansion

$$\log \mathbb{E} e^{tR} = t\mathbb{E}R + \frac{t^2}{2} \text{Var}(R) + O(t^3)$$

with $t = -1/\beta_k$ gives (30); boundedness makes the remainder uniform on the finite simplex.

A.3 PROOF OF THEOREM 1

Continuity on the compact policy simplex bounds $V_\beta(\pi)$, hence $\mathcal{G}_\beta$ is bounded. It is closed: if $u_n \rightarrow u$ and $u_n \leq V_\beta(\pi_n)$, compactness gives a subsequence $\pi_n \rightarrow \pi$, and continuity yields $u \leq V_\beta(\pi)$. Comprehensiveness follows immediately from the definition.

For convexity, take $u^i \leq V_\beta(\pi_i)$ for $i \in \{1, 2\}$ and $\lambda \in [0, 1]$. Objective-wise concavity gives

$$\lambda u^1 + (1 - \lambda)u^2 \leq \lambda V_\beta(\pi_1) + (1 - \lambda)V_\beta(\pi_2) \leq V_\beta(\lambda\pi_1 + (1 - \lambda)\pi_2).$$

Thus the convex combination lies in $\mathcal{G}_\beta$. Assumption 1 gives a point strictly above d , so the bargaining problem is nondegenerate.

The Nash objective is coordinatewise increasing. Hence an optimum over $\mathcal{G}_\beta$ can be moved to a dominating attainable vector $V_\beta(\pi)$ without decreasing its value, proving equivalence with (11).

The log objective tends to $-\infty$ when any surplus approaches zero. Since a strictly positive feasible surplus exists, the supremum is attained at a point strictly above disagreement. This proves existence and strict individual rationality.

If a policy π' weakly improved every game value over an optimizer π^* and strictly improved one, then every surplus would weakly increase and one would strictly increase. Strict monotonicity of the logarithm would imply $F(\pi') > F(\pi^*)$, a contradiction. Hence every optimizer is Pareto optimal.

For uniqueness of the utility outcome, suppose two maximizers π_1, π_2 induce distinct vectors. Concavity gives

$$V_\beta\left(\frac{1}{2}\pi_1 + \frac{1}{2}\pi_2\right) - d \geq \frac{1}{2}s(\pi_1) + \frac{1}{2}s(\pi_2)$$

coordinatewise. Since $G(s) = \sum_k \log s_k$ is increasing and strictly concave on $\mathbb{R}_{++}^K$,

$$\begin{aligned} F\left(\frac{1}{2}\pi_1 + \frac{1}{2}\pi_2\right) &\geq G\left(\frac{1}{2}s(\pi_1) + \frac{1}{2}s(\pi_2)\right) \\ &> \frac{1}{2}G(s(\pi_1)) + \frac{1}{2}G(s(\pi_2)), \end{aligned}$$

contradicting optimality. All maximizing policies therefore induce the same game-value vector.

A.4 PROOF OF THEOREM 2 AND PAYOFF RESCALING

Changing the utility coordinates together with their fallback values is distinct from changing the raw preference payoff while keeping the opponent temperature fixed. For $\lambda_k > 0$,

$$V_{\beta_k}^{\lambda_k A_k}(\pi) = \lambda_k V_{\beta_k/\lambda_k}^{A_k}(\pi). \quad (31)$$

Thus exact covariance at the oracle level requires jointly scaling (A_k, β_k) to $(\lambda_k A_k, \lambda_k \beta_k)$, or taking the unregularized limit.

Under $V'_k = c_k V_k + b_k$ and $d'_k = c_k d_k + b_k$, the transformed surplus is $s'_k = c_k s_k$. Therefore

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k,$$

a policy-independent shift. The optimizer set is unchanged.

For raw payoff scaling,

$$\begin{aligned} V_{\beta_k}^{\lambda_k A_k}(\pi) &= \min_{\nu} \{ \lambda_k g_k^A(\pi, \nu) + \beta_k \text{KL}(\nu \| \mu) \} \\ &= \lambda_k \min_{\nu} \{ g_k^A(\pi, \nu) + (\beta_k / \lambda_k) \text{KL}(\nu \| \mu) \}, \end{aligned}$$

which is (31). Setting the transformed temperature to $\lambda_k \beta_k$ gives $V_{\lambda_k \beta_k}^{\lambda_k A_k} = \lambda_k V_{\beta_k}^{A_k}$ and therefore exact Nash covariance.

A.5 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

Proposition 2 (Guarantees of competing rules). *There are jointly improvable finite pairwise games in which positive-weight utilitarian aggregation selects a policy below the fallback for one objective, and instances in which absolute maxmin violates fallback-relative individual rationality. Surplus maxmin is strictly individually rational under Assumption 1, but its optimizer set can contain a policy Pareto dominated by another optimizer. Every Nash optimizer has the guarantees in Theorem 1.*

A positive-weight utilitarian objective is strictly increasing in every coordinate, so every utilitarian maximizer is Pareto optimal. It need not be individually rational. Absolute maxmin can ignore objective-specific disagreement levels. Surplus maxmin is strictly individually rational under Assumption 1, because a feasible policy has positive minimum surplus, but its flat optimum can contain dominated ties. The following skew-symmetric games make all three failures explicit.

All examples use three responses and the uniform reference $\mu = (1/3, 1/3, 1/3)$. Utilities are unregularized game values $V_k(\pi) = \min_j (\pi^\top A_k)_j$.

Utilitarian failure of individual rationality. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

Then $d = (-1/3, -1/12)$. The policy $(1/4, 1/4, 1/2)$ has surplus $(1/12, 1/48) > 0$, but equal-weight utilitarian optimization has an optimizer $(0, 1/5, 4/5)$ with surplus $(7/30, -1/60)$.

Absolute-maxmin failure of individual rationality. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again $d = (-1/3, -1/12)$. The policy $(1/3, 7/15, 1/5)$ has surplus $(1/15, 1/15) > 0$. Absolute maxmin instead has an optimizer $(0, 4/5, 1/5)$ with utility $(-1/10, -1/10)$ and surplus $(7/30, -1/60)$.

A dominated surplus-maxmin tie. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

Here $d = (-1/3, -1/6)$. Objective 2 has maximal game value zero for every mixture supported on the first two responses, so no policy can have second surplus above $1/6$. Both

$$\pi_c = (1/3, 2/3, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$\pi_e = (0, 1, 0), \quad s(\pi_e) = (1/3, 1/6),$$

therefore maximize the minimum surplus at $1/6$, but π_e Pareto dominates π_c in game values.

A.6 PROOF OF THE REGRESSION CLAIM

Let

$$R_t(y, y') = \eta_t \sum_k \bar{\omega}_{t,k} \mathbb{E}_{z \sim \nu_{k,\pi_t}^*} [A_k(y, z) - A_k(y', z)].$$

The conditional mean of the binary target in (22) is $R_t(y, y')$. The squared-loss conditional-mean decomposition shows that any population minimizer must satisfy $h_t(\pi; y, y') = R_t(y, y')$ for every pair in the support of $\pi_t \otimes \pi_t$. The exponential mirror update satisfies this identity. Equality of all pairwise log differences implies that its log density and that of any other full-support minimizer differ only by an additive constant; normalization proves uniqueness.

A.7 PROOF OF THEOREM 3

For one prompt and one objective, write the payoff matrix as A_k . Equation (8) has Hessian

$$\nabla^2 V_{k,\beta_k}(\pi) = -\frac{1}{\beta_k} A_k \left(\text{Diag}(\nu_{k,\pi}^*) - \nu_{k,\pi}^* (\nu_{k,\pi}^*)^\top \right) A_k^\top. \quad (32)$$

For any direction h ,

$$-h^\top \nabla^2 V_{k,\beta_k}(\pi) h = \frac{1}{\beta_k} \text{Var}_{z \sim \nu_{k,\pi}^*} (h^\top A_k e_z) \leq \frac{1}{4\beta_k} \|h\|_1^2, \quad (33)$$

$$|h^\top \nabla V_{k,\beta_k}(\pi)| \leq \frac{1}{2} \|h\|_1. \quad (34)$$

On $s_k(\pi) \geq \varepsilon$,

$$-h^\top \nabla^2 \log s_k(\pi) h \leq \frac{1}{4} \left(\frac{1}{\beta_k \varepsilon} + \frac{1}{\varepsilon^2} \right) \|h\|_1^2. \quad (35)$$

Summing over objectives gives $L_\varepsilon \|h\|_1^2$. The Hessian of negative entropy $\psi(\pi) = \sum_y \pi(y) \log \pi(y)$ satisfies

$$h^\top \nabla^2 \psi(\pi) h = \sum_y \frac{h_y^2}{\pi_y} \geq \left(\sum_y |h_y| \right)^2 = \|h\|_1^2.$$

Integrating this curvature comparison along a line segment yields relative smoothness:

$$F(p) \geq F(q) + \langle \nabla F(q), p - q \rangle - L_\varepsilon \text{KL}(p \| q) \quad p, q \in \Pi_\varepsilon. \quad (36)$$

The exact mirror step satisfies the three-point inequality

$$\alpha_t \langle \nabla F(\pi_t), \pi - \pi_{t+1} \rangle \leq D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t), \quad (37)$$

where D is contextual KL. Relative smoothness and optimality of the update imply $F(\pi_{t+1}) \geq F(\pi_t)$ when $\alpha_t L_\varepsilon \leq 1$. Concavity of F , (36), and (37) with $\pi = \pi^*$ give

$$\alpha_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_t) - D(\pi^* \| \pi_{t+1}). \quad (38)$$

Summing telescopes. Since the objective gaps are nonincreasing,

$$\left(\sum_{t=0}^{T-1} \alpha_t \right) (F(\pi^*) - F(\pi_T)) \leq D(\pi^* \| \pi_0),$$

which proves (24) in the exact case $\xi_t = \delta_t = 0$.

It remains to translate welfare convergence into utility convergence. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\mu) \leq 0$ because $\nu = \mu$ is feasible and $g_k(\mu, \mu) = 0$. Hence every positive surplus satisfies $s_k \leq 1$. The function $G(s) = \sum_k \log s_k$ is therefore 1-strongly concave on the attainable positive surplus range. First-order optimality of the unique s^* over the convex comprehensive surplus set gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2.$$

Combining this inequality with the exact case of (24) proves the surplus-vector statement.

Approximate fitted updates. We say that the fitted policy has variational residual δ_t if, for every policy π ,

$$\alpha_t \langle \hat{g}_t, \pi - \pi_{t+1} \rangle \leq D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t) + \delta_t. \quad (39)$$

The exact mirror update has $\delta_t = 0$. Let $e_t = \hat{g}_t - \nabla F(\pi_t)$. Repeating the derivation of (38) with (39) gives

$$\begin{aligned} \alpha_t (F(\pi^*) - F(\pi_{t+1})) &\leq D(\pi^* \| \pi_t) - D(\pi^* \| \pi_{t+1}) \\ &\quad - \alpha_t \langle e_t, \pi^* - \pi_{t+1} \rangle + \delta_t. \end{aligned}$$

The contextual ℓ_1 distance between two policies is at most 2, so

$$-\langle e_t, \pi^* - \pi_{t+1} \rangle \leq 2\|e_t\|_\infty \leq 2\xi_t.$$

Therefore

$$\alpha_t(F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_t) - D(\pi^* \| \pi_{t+1}) + 2\alpha_t\xi_t + \delta_t. \quad (40)$$

Summing gives a weighted-average gap bound. If the accepted objective values are nondecreasing, the last gap is no larger than this weighted average, proving (24). Without monotonicity, the smallest gap among the iterates is no larger than the weighted average. The utility-vector statement follows from the same strong-concavity argument used above.

B FINITE-SAMPLE CERTIFICATION OF GAME VALUES

This section supplies the certificate invoked by Algorithm 1. It is deliberately conservative but fully explicit. Tighter empirical-Bernstein or paired bounds can replace it without changing the population objective.

B.1 NESTED ESTIMATOR

Fix at most R policy checks, including the cached reference check. Policies may be chosen adaptively, but check c uses a fresh, independent certification batch, or a sealed batch preallocated to that check before training and not revealed earlier. This independence is necessary; repeatedly reusing the same holdout after observing its outcomes is not covered by the theorem below.

For check c , draw n prompts $x_i \sim \rho$. For each prompt draw M comparators $z_{ij} \sim \mu(\cdot | x_i)$. For every objective, comparator, and prompt, draw L independent learner responses $y_{ij\ell} \sim \pi_c(\cdot | x_i)$ and binary outcomes

$$B_{k,c,i,j,\ell} \sim \text{Bernoulli}(P_k(y_{ij\ell} \succ z_{ij} | x_i)).$$

Define

$$\hat{r}_{k,c,i,j} = \frac{1}{L} \sum_{\ell=1}^L (B_{k,c,i,j,\ell} - \frac{1}{2}), \quad (41)$$

$$\hat{q}_{k,c,i} = \frac{1}{M} \sum_{j=1}^M \exp\{-\hat{r}_{k,c,i,j}/\beta_k\}. \quad (42)$$

Let

$$\epsilon_{\text{lab}} = \sqrt{\frac{\log(6KRnM/\delta)}{2L}}, \quad (43)$$

$$a_k = e^{-1/(2\beta_k)}, \quad b_k = e^{1/(2\beta_k)}, \quad (44)$$

$$\epsilon_{\text{cmp},k} = (b_k - a_k) \sqrt{\frac{\log(6KRn/\delta)}{2M}}, \quad (45)$$

$$\epsilon_{\text{pr}} = \sqrt{\frac{\log(6KR/\delta)}{2n}}. \quad (46)$$

For each sampled prompt define

$$\ell_{k,c,i} = -\beta_k \log \min \left\{ b_k, e^{\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} + \epsilon_{\text{cmp},k} \right\}, \quad (47)$$

$$u_{k,c,i} = -\beta_k \log \max \left\{ a_k, e^{-\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} - \epsilon_{\text{cmp},k} \right\}. \quad (48)$$

Finally set

$$\underline{V}_{k,c} = \max \left\{ -\frac{1}{2}, \frac{1}{n} \sum_{i=1}^n \ell_{k,c,i} - \epsilon_{\text{pr}} \right\}, \quad (49)$$

$$\overline{V}_{k,c} = \min \left\{ \frac{1}{2}, \frac{1}{n} \sum_{i=1}^n u_{k,c,i} + \epsilon_{\text{pr}} \right\}. \quad (50)$$

Theorem 5 (Simultaneous nested game-value intervals). *With probability at least $1 - \delta$, simultaneously for every objective $k \in [K]$ and every check $c \in [R]$,*

$$\underline{V}_{k,c} \leq V_{k,\beta_k}(\pi_c) \leq \bar{V}_{k,c}. \quad (51)$$

Consequently, if one check is the reference policy, the surplus interval in (16) is simultaneous and valid. If every lower surplus exceeds ε , then the checked policy belongs to Π_ε with probability at least $1 - \delta$.

B.2 PROOF OF THEOREM 5

For a fixed sampled comparator, $\hat{r}$ averages L independent variables in $[-1/2, 1/2]$. Hoeffding's inequality (Hoeffding, 1963) and a union bound over $KRnM$ estimates give

$$|\hat{r}_{k,c,i,j} - r_{k,\pi_c}(z_{ij} \mid x_i)| \leq \epsilon_{\text{lab}} \quad (52)$$

for all indices with failure probability at most $\delta/3$. The statement remains valid for adaptively selected π_c because the data used at check c are independent of the history conditional on which π_c is submitted for certification.

Let

$$q_{k,c,i} = \mathbb{E}_{z \sim \mu(\cdot \mid x_i)} \exp\{-r_{k,\pi_c}(z \mid x_i)/\beta_k\}$$

and let $q_{k,c,i}^M$ be its empirical mean over the M sampled comparators using the exact r values. Each exponential lies in $[a_k, b_k]$. A second Hoeffding bound and a union bound over KRn prompt-level means give

$$|q_{k,c,i}^M - q_{k,c,i}| \leq \epsilon_{\text{cmp},k} \quad (53)$$

with failure probability at most $\delta/3$.

On the intersection of (52) and (53),

$$e^{-\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} - \epsilon_{\text{cmp},k} \leq q_{k,c,i} \leq e^{\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} + \epsilon_{\text{cmp},k}.$$

Since $q_{k,c,i} \in [a_k, b_k]$ and $-\beta_k \log q$ is decreasing, (47) and (48) bound the exact prompt-level game value.

Each exact prompt-level value lies in $[-1/2, 1/2]$. A third Hoeffding bound over the n prompts, union-bounded over KR checks, gives prompt-generalization error at most ϵ_{pr} with failure probability at most $\delta/3$. Combining the three events proves (51). The surplus statement follows by subtracting the worst compatible endpoints of the candidate and reference intervals.

C CERTIFIED POSITIVE-SURPLUS WARM START

Define the population joint headroom

$$\gamma^* = \max_{\pi} \min_k s_k(\pi). \quad (54)$$

Assumption 1 is equivalent to $\gamma^* > 0$. The smooth objective in (14) satisfies

$$\min_k s_k(\pi) - \tau_0 \log K \leq H_{\tau_0}(\pi) \leq \max_k s_k(\pi). \quad (55)$$

Its gradient is

$$\nabla H_{\tau_0}(\pi) = \sum_{k=1}^K a_k(\pi) q_{k,\pi}, \quad a_k(\pi) = \frac{e^{-s_k(\pi)/\tau_0}}{\sum_j e^{-s_j(\pi)/\tau_0}}. \quad (56)$$

Thus Phase 0 uses the same regularized-opponent and mirror-regression machinery as Phase 1, but its objective weights remain finite at the reference. If $\tau_0 \log K < \gamma^*$, then the maximum of H_{τ_0} is positive.

The certificate is one-sided. An uncertified return does not establish $\gamma^* \leq 0$. A population infeasibility certificate would require an upper bound on (54), which is separate from finding a feasible warm start.

Algorithm 2 Certified Positive-Improvement Warm Start

Require: Reference μ , preference oracles $P_{1:K}$, temperatures $\beta_{1:K}$, floor ε , softmin temperature τ_0 , confidence δ , maximum checks R .

```
1: Initialize  $\pi^{(0)} \leftarrow \mu$  and preallocate independent certification batches.  
2: for  $r = 0, \dots, R - 1$  do  
3:   Certify  $\underline{s}_{r,k}$  for every objective using Appendix B.  
4:   if  $\underline{s}_{r,k} \geq 2\varepsilon$  for every  $k$  then  
5:     return  $\pi^{(r)}$ .  
6:   end if  
7:   Estimate softmin weights  $a_{r,k} \propto \exp\{-\widehat{s}_{r,k}/\tau_0\}$ .  
8:   Estimate each regularized opponent and form pairwise targets for  $\sum_k a_{r,k} q_{k,\pi^{(r)}}$ .  
9:   Fit  $\pi^{(r+1)}$  by the mirror-regression update.  
10: end for  
11: return “uncertified.”
```

Table 3: Exact optimization on the frozen ULTRAFEEDBACK response pool ($n = 1,372$ complete prompts). Surpluses use the population scale of Eq. (10); higher is better except for the Frank–Wolfe (FW) gap. This finite-pool certificate tests the bargaining rule, not neural generalization.

selection rule	min s_k	mean s_k	$\sum_k \log s_k$	FW gap
NBPO	0.221324	0.235957	-5.783311	9.02e-09
Utilitarian	0.217866	0.236070	-5.785865	9.86e-09

D FINITE-POOL POPULATION CERTIFICATE

We freeze 1,500 ULTRAFEEDBACK prompts, four policy responses and four reference responses per prompt, four objective-specific pairwise tensors, $\beta_k = 0.25$, and the reference game values before optimization. The complete intersection contains 1,372 prompts. We first obtain a positive-surplus Phase 0 point and then optimize (11) directly over one four-simplex per prompt in float64. The reported checkpoint is the converged final iterate, not a reward-selected iterate.

The Nash surplus vector is $(0.229230, 0.258926, 0.221324, 0.234350)$, so every coordinate is strictly above disagreement. Its normalized Frank–Wolfe gap is 9.02×10^{-9} ; the corresponding positive supporting weights certify that no policy can improve every game value by more than this residual. A deterministic feasible restart agrees in game-value space within 3.03×10^{-9} . Scaling the four payoff, temperature, and disagreement coordinates by $(0.2, 0.6, 1.0, 1.7)$ gives a scale-corrected surplus residual of 2.78×10^{-17} . The utilitarian control attains a slightly larger mean surplus, while the Nash solution attains the larger log product and minimum surplus. These calculations verify the finite-pool selection rule, but do not close the separate neural-regression error in Theorem 3.

Solution-concept stress test. We additionally preregister 200 new three-objective, five-action cyclic games, conditioned only on the joint-improvement assumption required by Theorem 1. We solve every population problem to numerical convergence and compare the resulting selection rules on the same game tensors. Table 4 separates the properties of the selection rule from neural approximation.

NBPO has zero Nash-welfare regret by construction, satisfies strict IR in every feasible game, and has maximum Pareto gap 4.36×10^{-13} . Under independent coordinate scaling by $(0.25, 0.60, 1.40)$, its maximum policy ℓ_1 drift is 7.01×10^{-7} , whereas PROSPER’s mean drift is 0.277. Surplus maxmin obtains the larger minimum surplus, as expected from its objective, while utilitarian obtains the larger mean surplus. The experiment therefore verifies the properties that distinguish the Nash rule; it is not evidence that one neural optimizer dominates every alternative metric.

E EXPERIMENTAL PROTOCOL AND DETAILED RESULTS

Run matrix. For each dataset and training run, construct one certified NBPO warm start and reuse it across the four game-value aggregation rules. Hold the prompt split, decoding parameters, candidate

Table 4: Exact population solutions on 200 preregistered cyclic games. “Defined” is the fraction for which the method returned a feasible policy, IR is the fraction strictly above the common fallback in every objective, and the Pareto gap is measured in the common game-value coordinates. MOPO uses objective 1 as primary and preregistered half-attainable thresholds. Methods optimize different social objectives, so the minimum surplus is diagnostic rather than a universal ranking.

selection rule	defined	strict IR	mean min. surplus	Pareto gap
NBPO	1.000	1.000	0.0289	1.64×10^{-13}
PROSPER/MaxEntBW	1.000	0.665	0.0071	1.86×10^{-13}
Utilitarian	1.000	0.595	-0.0009	1.42×10^{-13}
Surplus maxmin	1.000	1.000	0.0361	2.16×10^{-13}
MOPO	0.510	0.035	-0.0887	2.67×10^{-2}
HT single-objective corners	1.000	0.030–0.060	-0.1351–-0.1207	$5.95\text{--}6.75 \times 10^{-13}$

count, comparator pool, labeled pair set, total preference-query budget, total optimizer budget, and $\beta_{1:K}$ fixed within a seed. Count Phase 0, opponent scoring, rejected stages, and validation used for hyperparameter selection. PROSPER, fixed-reference precursor, MOPO, and Uniform-INPO are external baselines and should retain their canonical update rules while receiving the same total resource budget.

Independent evaluation. Training labels, utility certification, and final evaluation must use disjoint data. The final game values use an independently generated comparator pool and evaluator families that did not provide training labels. Report failure to certify joint positive surplus as a result rather than replacing it with a clipped Phase 1 run.

Primary metrics. For a final release, report V_{k,β_k} and s_k on the theoretical scale of (2), together with the minimum surplus, average game value, Nash welfare, IR rate, fixed-reference win rate, KL to the reference, response length, opponent entropy, and accepted-stage rate. The currently archived summary provides only the implementation-scale quantities $\tilde{s}_k$ and $\tilde{F}$ shown below. Rows without a canonical implementation audit are omitted. Nash welfare is undefined when a held-out surplus is nonpositive; do not apply an unreported offset.

E.1 FROZEN REPORTING TABLES FOR THE MATCHED NEURAL STUDY

The following tables fix the reporting surface for the new matched-budget study before its final evaluation. A dash denotes a cell that has not yet been measured; it must not be imputed from a different model, judge, prompt split, or training budget. The primary panel uses four global criteria on held-out UltraFeedback prompts: instruction following, truthfulness, honesty/calibration, and helpfulness. All methods share one response pool and comparison tensor. PROSPER retains its alternating objective-weight and opponent updates, while MOPO uses instruction following as the preregistered primary objective with fixed lower bounds on the other three objectives. The four HT-MNPO rows are single-objective corner policies rather than candidate compromise policies.

These tables serve different claims. Table 6 tests whether the learned compromise approaches the intended bargaining outcome under genuine objective conflict. Table 2 checks that gains are not caused by verbosity or broad capability regression. Table 7 tests whether the result transfers to safety and summarization rather than depending on one prompt distribution.

Remaining reporting checks. A final multi-seed evaluation should document the payoff normalization, plot each run’s theoretical minimum surplus and Nash welfare across stages, mark Phase 0 and rejected steps, test opponent-pool sensitivity, audit generic or excessive-refusal behavior, and report both game values and conventional fixed-reference win rates.

F ULTRAFEEDBACK-SCALE MATCHED PROJECTION STUDY

This study instantiates the matched-comparison protocol of Appendix E on 7,000 ULTRAFEEDBACK prompts with four objectives (instruction following, truthfulness, honesty, helpfulness), a pinned

Table 5: Main comparison from the available completed matched run, on the population scale of (2). “min s ” is the smallest objective improvement, $F = \sum_k \log s_k$, and “ref. win” is the average conventional win rate against the reference. Higher is better within a panel. Detailed objective values are in Appendix E. PROSPER and MOPO are omitted pending evaluation with their canonical algorithms.

method	HELPSTEER2			SAFEERLHF		
	min s	F	ref. win	min s	F	ref. win
NBPO (ours)	0.104	-8.011	0.537	0.500	-1.319	0.817
Utilitarian	0.109	-7.931	0.560	0.510	-1.314	0.821
Absolute maxmin	0.129	-6.978	0.577	0.496	-1.325	0.811
Surplus maxmin	0.133	-7.286	0.575	0.497	-1.332	0.817
Fixed-reference precursor	0.134	-7.251	0.565	0.520	-1.278	0.837
Uniform-INPO	0.079	-9.156	0.514	0.495	-1.328	0.804

Table 6: Improvement comparison table. Entries are held-out surplus estimates on the scale of (2). IR is the fraction of prompts for which all four lower confidence bounds exceed the fallback. Completed rows use one 663-prompt jointly complete held-out panel judged in a single pass; “ref. win” is the mean anchored win rate. All trained arms use the manuscript loss (22) run open-loop, without the per-stage surplus certificate of Algorithm 1. At $4\times$ the manuscript’s optimizer budget every rule collapses (mean response length nearly doubles, anchored win rate falls below 0.09); at the manuscript budget the policies remain fluent and the rules separate, but none recovers the certified warm start’s individual rationality. “IR” is the fraction of held-out prompts on which all four objectives exceed the fallback; the base policy scores 0.248 on this statistic, so it is the sampling floor rather than zero. Paired 2,000-resample prompt-bootstrap 95% intervals for the minimum surplus are $[+0.005, +0.033]$ (MOPO $\tau=0.1$), $[-0.014, +0.014]$ (MOPO $\tau=1$), $[-0.029, +0.008]$ (Uniform-INPO), $[-0.077, -0.044]$ (NBPO) and $[-0.092, -0.059]$ (corner helpfulness). MOPO at $\tau=0.1$ is the only arm whose interval is strictly positive; the single-objective corners and every aggregation rule other than surplus maxmin and Uniform-INPO are significantly negative. Its importance weights are extremely concentrated—clipping and normalising leave an effective sample size of 879 of 24,000 pairs, against 9,799 at $\tau=1$ —so it trains on a small high-scoring subset rather than the full pair set. Remaining rows require baselines that have not yet been trained on this panel.

method	instr. s	truth. s	honest s	helpful s	min s	IR	ref. win
Certified warm start (Phase 0)	-0.0011	0.0073	0.0101	0.0161	-0.0011	0.270	0.502
<i>Manuscript budget (300 steps)</i>							
MOPO ($\tau=0.1$, instruction primary)	0.0289	0.0253	0.0220	0.0296	0.0220	0.297	0.519
MOPO ($\tau=1$, instruction primary)	0.0187	0.0073	0.0042	0.0042	0.0042	0.259	0.502
<i>Manuscript budget (300 steps)</i>							
NBPO (ours)	-0.0542	-0.0583	-0.0477	0.0336	-0.0583	0.226	0.450
NBPO, prompt-wise	-0.0379	-0.0597	-0.0446	0.0387	-0.0597	0.220	0.454
Utilitarian	-0.0669	-0.0803	-0.0713	0.0238	-0.0803	0.219	0.430
Absolute maxmin	-0.0451	-0.0640	-0.0521	0.0556	-0.0640	0.255	0.453
Surplus maxmin	-0.0138	-0.0013	0.0222	0.1048	-0.0138	0.297	0.515
PROSPER/MaxEntBW	-0.0364	-0.0502	-0.0455	0.0648	-0.0502	0.237	0.468
<i>Uncertified $4\times$ budget (1,200 steps)</i>							
NBPO	-0.3607	-0.3755	-0.3862	-0.3531	-0.3862	0.006	0.071
Utilitarian	-0.3382	-0.3587	-0.3689	-0.3275	-0.3689	0.033	0.093
Uniform-INPO	-0.0110	0.0222	0.0226	0.0783	-0.0110	0.299	0.517
Single-objective corner (truthfulness)	-0.0489	-0.0477	-0.0388	0.0234	-0.0489	0.237	0.455
Single-objective corner (helpfulness)	-0.0608	-0.0758	-0.0693	0.0463	-0.0758	0.205	0.442

Qwen3-32B guided-choice judge queried in both presentation orders, four candidate responses and four reference anchors per prompt (896,000 judgments per side), and a 6,000/1,000 train/held-out prompt split. All aggregation rules share one certified positive-surplus parent policy, one frozen pair set, and one optimizer budget.

Table 7: Frozen cross-domain reporting table. SafeRLHF uses helpfulness and harmlessness pairwise judges. TL;DR uses factuality, coverage, and conciseness criteria on a disjoint Reddit-summary split. Every comparison is made within a dataset and protocol; scores are never normalized jointly across the two panels. The completed table will report paired confidence intervals, exact evaluator revisions, prompt counts, and training seeds.

method	SafeRLHF			TL;DR			
	helpful s	harmless s	min s	factual s	coverage s	concise s	min s
NBPO (ours)	–	–	–	–	–	–	–
PROSPER/MaxEntBW	–	–	–	–	–	–	–
Utilitarian	–	–	–	–	–	–	–
Absolute maxmin	–	–	–	–	–	–	–
Surplus maxmin	–	–	–	–	–	–	–
Uniform-INPO	–	–	–	–	–	–	–
MOPO	–	–	–	–	–	–	–
HT-MNPO corners	–	–	–	–	–	–	–

Table 8: Detailed finite- β HELPSTEER2 results from the available matched run, on the population scale of (2). “IR” is the number of runs with positive improvement on every objective.

method	helpful s	correct s	coherent s	concise s	min s	F	IR
NBPO	0.197	0.134	0.104	0.121	0.104	−8.011	1/1
Utilitarian	0.199	0.132	0.109	0.125	0.109	−7.931	1/1
Absolute maxmin	0.221	0.164	0.129	0.198	0.129	−6.978	1/1
Surplus maxmin	0.233	0.159	0.133	0.139	0.133	−7.286	1/1
Fixed-reference precursor	0.223	0.154	0.134	0.154	0.134	−7.251	1/1
Uniform-INPO	0.170	0.099	0.080	0.079	0.079	−9.156	1/1

Target reliability across independent candidate halves. Table 12 reports the split-half reproducibility of each rule’s centered per-response target: the target is computed twice from disjoint two-candidate halves, and the two versions are correlated over all held-out responses with a paired 2,000-resample prompt bootstrap. Every objective has strictly positive policy-minus-reference surplus in each half. The NBPO target is significantly more reproducible than both maxmin targets and statistically indistinguishable in practical terms from the utilitarian target, consistent with the near-uniform inverse-surplus weights that a balanced panel induces.

Held-out realization of the regression target. Table 13 measures whether the trained network reproduces its prescribed mirror target on held-out prompts: the pairwise policy log-ratio is correlated against the frozen centered target on 148 held-out prompt groups after training on 1,340 groups for exactly four passes. Same-split memorization unit tests (32 groups, 50 passes) exceed Pearson 0.98 for every configuration shown, so optimizer and capacity are not the binding constraint; the target itself has split-half reliability above 0.88, so label noise is not the ceiling either. Across the logistic and squared losses, LoRA ranks 64 and 256, and a full-precision forward pass, held-out realization plateaus near 0.36, far below the 0.80 preregistered for opening reward-based selection. We are not

Table 9: Detailed finite- β SAFERLHF results from the available matched run, on the population scale of (2). “IR” is the number of runs with positive improvement on every objective.

method	helpful s	harmless s	min s	F	IR
NBPO	0.500	0.534	0.500	−1.319	1/1
Utilitarian	0.510	0.527	0.510	−1.314	1/1
Absolute maxmin	0.496	0.536	0.496	−1.325	1/1
Surplus maxmin	0.497	0.531	0.497	−1.332	1/1
Fixed-reference precursor	0.520	0.535	0.520	−1.278	1/1
Uniform-INPO	0.495	0.535	0.495	−1.328	1/1

Table 10: Temperature and certification ablation on the population scale of (2). “Gap” is the reported difference from the fixed-reference precursor. It should not be compared numerically with Theorem 4 until the reporting-scale normalization is documented.

β	min s	F	reported gap	opp. entropy	accepted stages	ref. win
∞	0.520	-1.278	0	n/a	2/2	0.837
1.0	0.487	-1.369	-0.034	0.988	2/2	0.803
0.25	0.500	-1.319	-0.020	0.877	2/2	0.817
0.05	0.475	-1.385	-0.046	0.592	2/2	0.802
0 (finite game only)	0.569	-1.078	-0.019	0	n/a	n/a

Table 11: Raw-label scaling intervention. The coupled condition scales both A_k and β_k by λ and should preserve the population target; the fixed-temperature condition intentionally changes the comparator neighborhood.

condition	λ	target cosine	policy shift	min $\tilde{s}$	opp. entropy
coupled (A_k, β_k)	0.6	1.000000	0.195	0.600	0.877
coupled (A_k, β_k)	0.3	1.000000	0.471	0.300	0.877
fixed β_k	0.6	0.993629	0.273	0.751	0.941
fixed β_k	0.3	0.972431	0.476	0.556	0.983

Table 12: Split-half reliability of the centered selection targets on 7,000 ULTRAFEDBACK prompts. “argmax” is the fraction of prompts whose top-scored response agrees between halves. Δ is NBPO minus the row’s rule (paired bootstrap 95% interval).

selection rule	split-half Pearson [95% CI]	argmax	Δ vs. NBPO [95% CI]
NBPO	0.884 [0.879, 0.888]	0.787	—
Utilitarian	0.894 [0.890, 0.897]	0.798	-0.010 [-0.012, -0.009]
Absolute maxmin	0.869 [0.864, 0.875]	0.932	+0.014 [+0.008, +0.020]
Surplus maxmin	0.639 [0.629, 0.649]	0.617	+0.244 [+0.236, +0.253]

aware of prior preference-optimization work that reports this quantity; the plateau motivates the pair-oriented projection used in Table 6.

Table 13: Held-out realization of the prescribed regression target (ULTRAFeEDBACK, 1,340 train / 148 held-out prompt groups, four passes, best learning rate per grid, fp32 forward). “KL” is the median per-prompt KL from the parent.

projection	adapter	train Pearson	held-out Pearson	held-out KL
Logistic (Bradley–Terry)	LoRA-64	0.954	0.366	0.018
Squared (22)	LoRA-64	0.953	0.366	0.018
Squared (22)	LoRA-256	0.928	0.363	0.018

Independence from objective utility scales. Theorem 2 states that the NBPO solution is unchanged when each objective’s utility is rescaled independently. Table 14 tests this on the frozen pool without retraining or relabelling: each objective’s payoff and temperature are multiplied by its own factor λ_k , which leaves the underlying preferences untouched, and every rule’s selection is recomputed. We report the fraction of the $6,000 \times 6$ response pairs whose preferred element changes. NBPO does not move a single pair under any of the three rescalings, matching the theorem exactly. Every competing rule does: the utilitarian average and PROSPER’s per-prompt worst objective both compare raw values across objectives, and the two maxmin rules move most of all. The zeros in the maxmin rows are coincidental rather than structural—they occur only when a rescaling happens to leave the arg-minimising objective unchanged—whereas the NBPO zeros hold for every rescaling. The prompt-wise NBPO variant is close to invariant but not exactly so, because a prompt whose estimated surplus is nonpositive needs a numerical floor and the floor does not commute with rescaling; the same floor is what makes the panel-level rule deviate when it binds.

Table 14: Independent objective rescaling on the frozen 7,000-prompt ULTRAFeEDBACK pool. Each entry is the percentage of response pairs whose preferred element flips when objective k ’s payoff and temperature are both multiplied by λ_k , a transformation that leaves every underlying preference unchanged. Lower is better; 0 is the value implied by Theorem 2. No training, decoding, or judging is repeated.

selection rule	$\lambda^{(A)}$	$\lambda^{(B)}$	$\lambda^{(C)}$	mean
NBPO (ours)	0.00	0.00	0.00	0.00
NBPO, prompt-wise	2.53	2.97	2.32	2.61
Utilitarian	5.66	6.31	5.85	5.94
PROSPER/MaxEntBW	7.35	6.63	6.71	6.90
Surplus maxmin	16.02	13.57	0.00	9.86
Absolute maxmin	16.32	0.00	16.32	10.88

The rescaling vectors, in the objective order (helpfulness, honesty, instruction following, truthfulness), are $\lambda^{(A)} = (0.2, 0.6, 1.0, 1.7)$, $\lambda^{(B)} = (3.0, 0.5, 1.0, 0.25)$ and $\lambda^{(C)} = (1.0, 1.0, 1.0, 5.0)$.

Matched game-value comparison. Table 6 in Appendix E.1 evaluates all arms on the 1,000 held-out prompts under the frozen evaluator, anchors, and decoding shared with the parent run. “pair-projection” arms orient each frozen response pair by the rule’s selector score and train with a standard length-normalized preference loss; “exact regression” arms regress the pairwise log-ratio difference onto the rule’s signed score difference with the manuscript loss (22). Both use the identical 1,200-step budget. The completed rows stress-test the certificate requirement of Algorithm 1. Run open-loop at four times the manuscript’s optimizer budget, the squared-loss update drives every rule to the same collapsed policy: mean response length grows from 1,399 to about 2,700 characters, the anchored win rate falls to 0.07–0.09, and every held-out surplus is strongly negative. Restoring the manuscript’s own 300-step budget removes the collapse—response lengths stay within $1.2\text{--}1.3\times$ the parent and generations remain fluent—and the aggregation rules then separate measurably. NBPO dominates the utilitarian rule on every objective, on the minimum surplus, and on the anchored win rate, which is the comparison the inverse-surplus weights are designed to affect; both maxmin rules place ahead of NBPO on this panel, with surplus maxmin closest to the fallback. Critically, no trained

Table 15: Four-objective HELPSTEER2 fixed-reference precursor results with a prompted pairwise oracle. Entries report win rates against the reference, averaged over Phi-4 and Llama-3.3-70B, neither of which supplied training preferences. The upper block reports four-stage runs, while the lower block reports single-stage baselines for context. Results are from one training seed.

rule	helpful	correct	coherent	concise	Avg	Worst
<i>four stages each</i>						
NBPO-KL	0.660	0.642	0.555	0.471	0.582	0.471
Utilitarian	0.625	0.597	0.530	0.426	0.545	0.426
MaxMin	0.463	0.424	0.382	0.369	0.410	0.369
<i>single stage</i>						
Utilitarian, 1 stage	0.599	0.583	0.559	0.416	0.539	0.416
MaxMin, 1 stage	0.604	0.587	0.559	0.420	0.543	0.420
HT-MNPO-helpful	0.654	0.618	0.593	0.324	0.547	0.324
HT-MNPO-correct	0.629	0.617	0.574	0.381	0.550	0.381
HT-MNPO-coherent	0.631	0.596	0.579	0.351	0.539	0.351
HT-MNPO-concise	0.422	0.470	0.448	0.722	0.515	0.422
INPO	0.616	0.604	0.560	0.425	0.551	0.425
SimPO	0.618	0.610	0.560	0.429	0.554	0.429
IPO	0.609	0.589	0.553	0.424	0.544	0.424
DPO	0.603	0.574	0.561	0.417	0.539	0.417
Base (μ)	0.500	0.500	0.500	0.500	0.500	0.500

Table 16: Three training seeds of the two four-stage fixed-reference rules on HELPSTEER2. Scored by Phi-4 on 500 held-out prompts. The $\pm$ figures are standard deviations over training seeds. The ordering does not persist across seeds.

rule	seed 42	seed 43	seed 44	mean
<i>worst objective</i>				
NBPO-KL	0.504	0.398	0.328	0.410 $\pm$.088
Utilitarian	0.461	0.490	0.318	0.423 $\pm$.092
<i>average objective</i>				
NBPO-KL	0.593	0.449	0.385	0.475 $\pm$.106
Utilitarian	0.551	0.601	0.373	0.508 $\pm$.120

arm at either budget attains joint positive surplus, whereas the certified warm start does. Since the same collapse appears under every rule, it is attributable to the uncertified projection step rather than to the aggregation rule, and it is consistent with the held-out realization plateau of Table 13: when the fitted update does not track its prescribed target off-distribution, the accepted-stage certificate is the mechanism that prevents drift, exactly as the population analysis assumes. We therefore report these runs as a negative result about uncertified projection, not as evidence for or against the bargaining rule.

G FIXED-REFERENCE PRECURSOR EXPERIMENTS

The tables below preserve the numerical results from the preceding NBPO draft. These runs use the anchor margin $a_k(\pi) = P_k(\pi \succ \mu) - 1/2$, add $-0.05\text{KL}(\pi\|\mu)$ to the welfare objective, initialize directly at μ , and use clipped inverse-surplus weights. By Theorem 4, they concern the $\beta = \infty$ endpoint, not finite-temperature NBPO. They are retained as an audit trail and must not be cited as evidence for the regularized-opponent objective.

Population optimizer audit. The full and minimal exact-gradient updates were identical, and absorbing the sum of inverse-surplus weights into the step size changed the policy by at most $2.9e - 15$ in ℓ_1 . Removing Phase 0 increased Nash-welfare regret by 0.0187, while a stage-level acceptance test reduced regret under frozen gradient noise by 0.0372. Raw inverse weights at the same nominal step reduced the strict-IR rate by 4.0%. The scale-covariance and Pareto residuals were below $2.4e - 15$.

Table 17: SAFERLHF fixed-reference precursor results. Entries are win rates against Llama-3.1-8B-Instruct on 500 held-out prompts, averaged over Phi-4 and Llama-3.3-70B. The subscript on NBPO-KL is the spread over three training seeds.

rule	helpful	harmless	Avg	Worst
NBPO-KL	0.807	0.845	0.826	$0.807_{\pm .014}$
Utilitarian	0.816	0.845	0.830	0.816
MaxMin	0.806	0.853	0.830	0.806
HT-MNPO-helpful	0.803	0.784	0.794	0.784
HT-MNPO-harmless	0.769	0.853	0.811	0.769
INPO	0.808	0.816	0.812	0.808
SimPO	0.780	0.764	0.772	0.764
IPO	0.753	0.727	0.740	0.727
DPO	0.704	0.716	0.710	0.704
Base (μ)	0.500	0.500	0.500	0.500

Table 18: Fixed-reference target under symmetric coin-flip contamination of the helpfulness judge on HELPSTEER2. The exact proportionality is a property of the $\beta = \infty$ anchor limit. It is not the prediction for finite-temperature NBPO with fixed β .

rule	λ	ω helpful	ω concise	mean $ t $	cosine	ratio spread
NBPO-KL	1.0	0.076	0.470	0.393	n/a	n/a
NBPO-KL	0.6	0.120	0.447	0.374	1.000000	8×10^{-15}
NBPO-KL	0.3	0.214	0.399	0.334	1.000000	5×10^{-15}
Utilitarian	1.0	0.250	0.250	0.403	n/a	n/a
Utilitarian	0.6	0.250	0.250	0.363	0.999105	1.28
Utilitarian	0.3	0.250	0.250	0.332	0.996738	2.24
MaxMin	any	0.000	1.000	0.411	1	0

These results support a minimal population update: retain the feasibility warm start, normalize inverse-surplus weights or absorb their sum into the step size, and omit post-warm-start clipping. Stage acceptance remains an outer safeguard for inexact neural updates.

The simultaneous Hoeffding lower bound made no false certifications in this study, compared with 1.09% for the plug-in rule at $n = 128$, but its power was only 9.0% at $n = 8192$. We therefore use confidence bounds as external certificates rather than as an optimization ingredient.

Interpretation. The precursor runs provide no reliable finite- β comparison. The three-seed HELPSTEER2 ordering changes sign, and repeated fixed-reference updates can leave the positive-surplus region, especially without normalized weights. These observations motivate matched seed-level evaluation, a certified warm start, and stage-wise reporting, but they are not performance claims for NBPO.

Table 19: Fixed-reference ablation of weight normalization and pair selection on SAFERLHF. Entries are reference win rates on 500 held-out prompts. The sharp degradation without normalization motivates treating the normalized coefficient as an explicit effective step size.

weights	pairs	stage	Qwen3-32B		Phi-4	
			Avg	Worst	Avg	Worst
normalized	best-of-4	1	0.737	0.722	0.772	0.747
normalized	best-of-4	2	0.698	0.684	0.724	0.710
normalized	random	1	0.730	0.699	0.745	0.707
normalized	random	2	0.667	0.635	0.665	0.608
unnormalized	best-of-4	1	0.490	0.457	0.499	0.490
unnormalized	best-of-4	2	0.095	0.045	0.032	0.028
unnormalized	random	1	0.510	0.478	0.508	0.503
unnormalized	random	2	0.380	0.349	0.363	0.355

Table 20: Preregistered population-optimizer audit over 200 finite cyclic games. Regret differences are paired means with 95% bootstrap intervals.

comparison	endpoint	result
full vs. minimal	max policy ℓ_1	$0.0e + 00$
normalized vs. absorbed step	max policy ℓ_1	$2.9e - 15$
without Phase 0	Nash-regret increase	0.0187 [0.0165, 0.0213]
noisy stage gate	Nash-regret change	-0.0372 [-0.0407, -0.0340]
raw fixed step	strict-IR change	-0.040 [-0.070, -0.015]

H JUDGE INSTANTIATION AND PREFERENCE LABELING

Training judges. Use the same prompted pairwise interface on both panels. For each objective, Qwen3-32B compares two responses under an attribute-specific rubric. Each pair is queried in both presentation orders, and the verdicts are swap-averaged. The fixed-reference precursor also contains deterministic Beaver reward and cost heads; those runs should not be conflated with the prompted-oracle main comparison.

Evaluation judges. Evaluation is separate from training and uses Phi-4 and Llama-3.3-70B. Each evaluator returns `[[A]]`, `[[B]]`, or `[[TIE]]`, scored as 1, 0, or 1/2, in both presentation orders. This makes the empirical evaluation oracle skew-symmetric after averaging.

Rubrics. Helpfulness asks which response better accomplishes the user’s stated goal; correctness asks which is more accurate; coherence asks which is clearer and better organized; conciseness asks which conveys the needed content with less padding; and harmlessness asks which is safer. The judge is instructed to evaluate only the designated objective.

Prompt template. For concreteness, the helpfulness system message and pairwise user message are

```
System: You compare two AI responses on HELPFULNESS ONLY:
which better helps the user accomplish their stated goal
(relevance, substance, actionable detail). Ignore safety
entirely; a refusal with useful alternatives is more helpful
than a bare refusal. End with exactly [[A]] or [[B]] or
[[TIE]].
User: [User prompt] {prompt} [Response A] {A} [Response B]
{B} Verdict:
```

The other objectives use the same template with only the rubric changed, and each pair is queried again after swapping responses A and B.

From verdicts to regularized opponents. For a current policy π_t and a reference comparator z , estimate

$$\hat{r}_{k,\pi_t}(z | x) = \frac{1}{L} \sum_{\ell=1}^L (B_k(y_\ell, z) - \frac{1}{2}), \quad y_\ell \sim \pi_t(\cdot | x).$$

Reference responses are reweighted by $\exp\{-\hat{r}_{k,\pi_t}(z | x)/\beta_k\}$ and normalized to sample $z_k \sim \hat{\mathcal{V}}_{k,\pi_t}^*$. For a policy pair (y, y') sharing z_k , the binary difference in (20) enters the regression target directly. No scalar reward model or value function is fitted by NBPO.